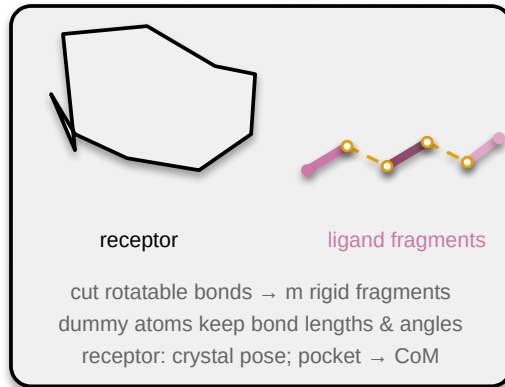
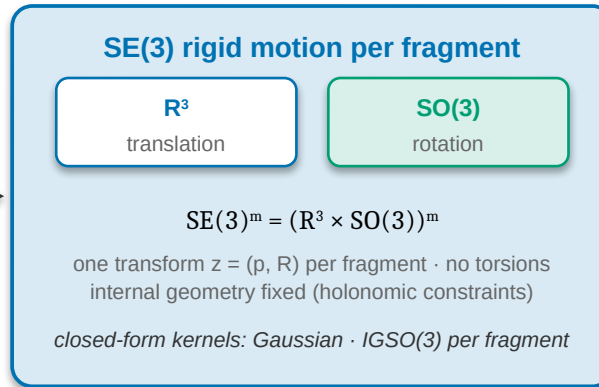


SigmaDock: fragment-based SE(3) diffusion docking

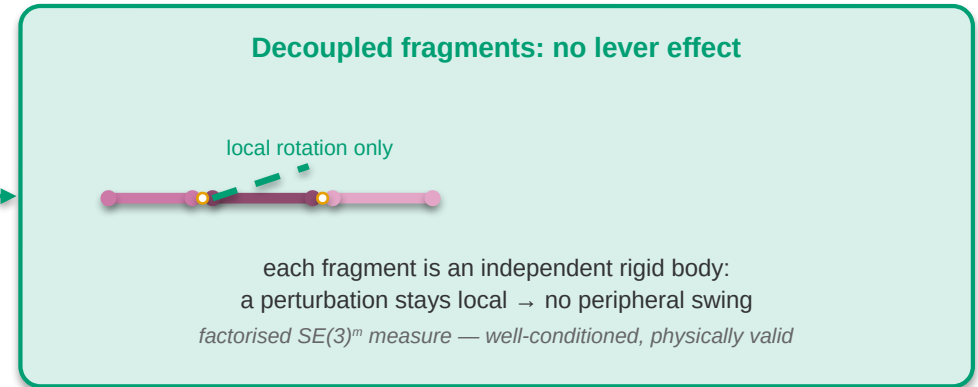
INPUTS



POSE REPRESENTATION



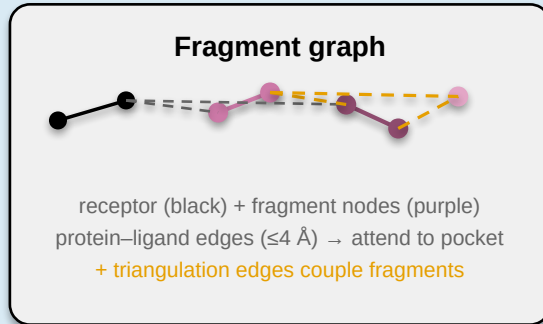
DECOUPLED PARAMETERISATION



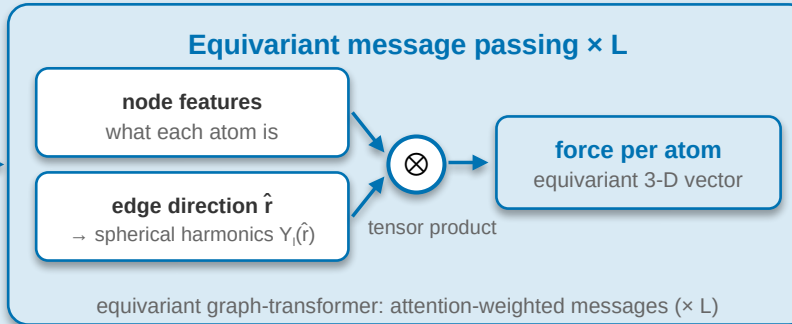
SE(3)-equivariant score model $s_\theta(z, t)$ — EquiformerV2

evaluated at each noise level t ($\times 25$ steps)

Fragment graph

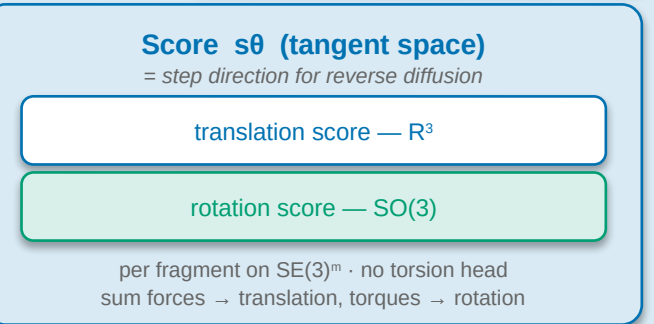


Equivariant message passing $\times L$



Score s_θ (tangent space)

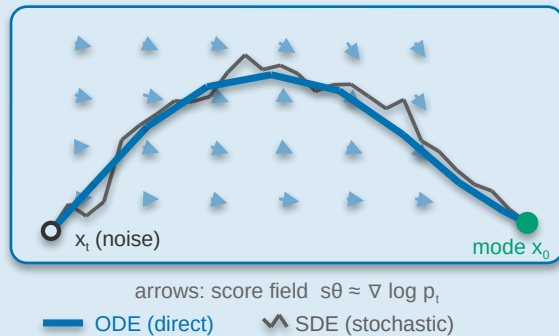
= step direction for reverse diffusion



SAMPLING

Reverse diffusion

deterministic ODE, 25 steps, $\eta \rightarrow 0$ (quadratic anneal)



N seeds

candidate poses



an ensemble

Ranking heuristic

no learned confidence model

Vinardo energy $\times$ PoseBusters

$$s_i = -b / p_i^b \rightarrow \text{rank, top-1}$$

cheap, trivially parallel,
model-agnostic

RANKED POSES IN POCKET

